

☒ EASY

☐ ADVANCED MATH

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# Advanced Math

# 13

10 questions · ~15 min

**Q1**

ADVANCED MATH

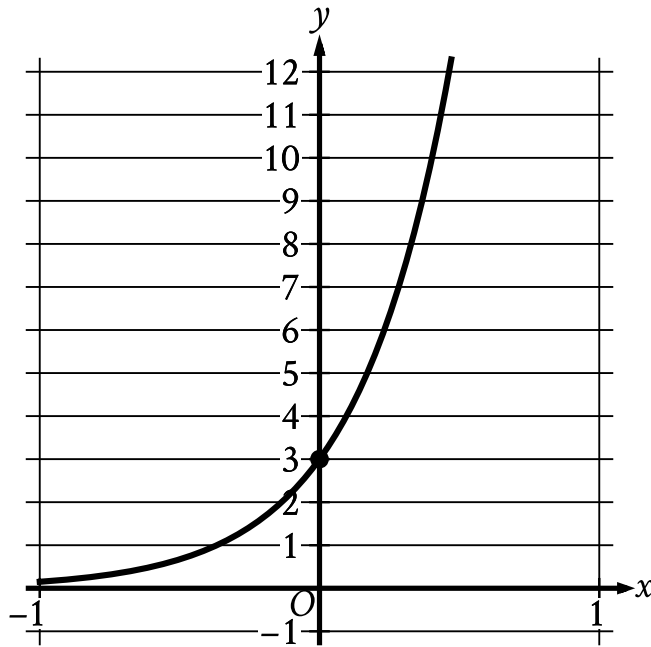
Which expression is equivalent to  $50x^2 + 5x^2$ ?

A.  $250x^2$

B.  $10x^2$

C.  $45x^2$

D.  $55x^2$



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up sharply to the point (0 comma 3).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - approximately (negative 1 comma 0.2)
  - (0 comma 3)

The graph of the exponential function  $f$  is shown, where  $y = f(x)$ . The  $y$ -intercept of the graph is  $(0, y)$ . What is the value of  $y$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
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|                             |                             |                             |                             |
|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| $\frac{1}{\textcircled{2}}$ | $\frac{1}{\textcircled{2}}$ | $\frac{1}{\textcircled{2}}$ | $\frac{1}{\textcircled{2}}$ |
| $\textcircled{3}$           | $\textcircled{3}$           | $\textcircled{3}$           | $\textcircled{3}$           |
| $\textcircled{4}$           | $\textcircled{4}$           | $\textcircled{4}$           | $\textcircled{4}$           |
| $\textcircled{5}$           | $\textcircled{5}$           | $\textcircled{5}$           | $\textcircled{5}$           |
| $\textcircled{6}$           | $\textcircled{6}$           | $\textcircled{6}$           | $\textcircled{6}$           |
| $\textcircled{7}$           | $\textcircled{7}$           | $\textcircled{7}$           | $\textcircled{7}$           |
| $\textcircled{8}$           | $\textcircled{8}$           | $\textcircled{8}$           | $\textcircled{8}$           |
| $\textcircled{9}$           | $\textcircled{9}$           | $\textcircled{9}$           | $\textcircled{9}$           |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q3**

ADVANCED MATH

$$8j = k + 15m$$

The given equation relates the distinct positive numbers  $j$ ,  $k$ , and  $m$ . Which equation correctly expresses  $j$  in terms of  $k$  and  $m$ ?

- A.  $j = \frac{k}{8} + 15m$
- B.  $j = k + \frac{15m}{8}$
- C.  $j = 8(k + 15m)$
- D.  $j = \frac{k + 15m}{8}$

Q4

ADVANCED MATH

$$f(x) = 2(3^x)$$

For the function  $f$  defined above, what is the value of  $f(2)$ ?

- A. 9
- B. 12
- C. 18
- D. 36

Q5

ADVANCED MATH

$$p + 34 = q + r$$

The given equation relates the variables  $p$ ,  $q$ , and  $r$ . Which equation correctly expresses  $p$  in terms of  $q$  and  $r$ ?

- A.  $p = q + r + 34$
- B.  $p = q + r - 34$
- C.  $p = -q - r + 34$
- D.  $p = -q - r - 34$

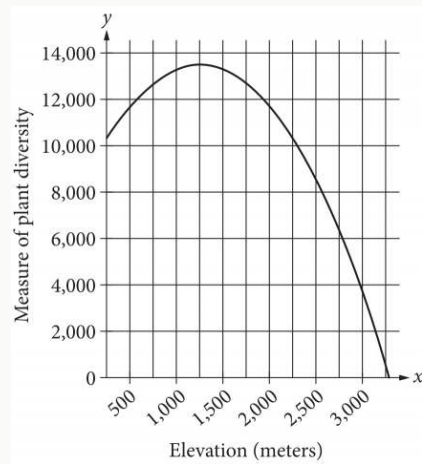
A ball is dropped from an initial height of 22 feet and bounces off the ground repeatedly. The function  $h$  estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. Which equation defines  $h$ , where  $h(n)$  is the estimated maximum height of the ball after it has hit the ground  $n$  times and  $n$  is a whole number greater than 1 and less than 10?

A.  $h(n) = 22(0.22)^n$

B.  $h(n) = 22(0.85)^n$

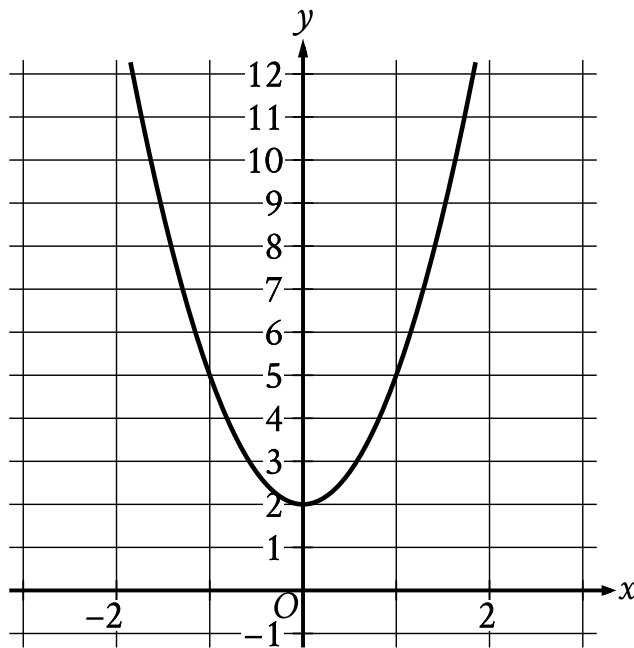
C.  $h(n) = 850.22^n$

D.  $h(n) = 85(0.85)^n$



The quadratic function graphed above models a particular measure of plant diversity as a function of the elevation in a region of Switzerland. According to the model, which of the following is closest to the elevation, in meters, at which plant diversity is greatest?

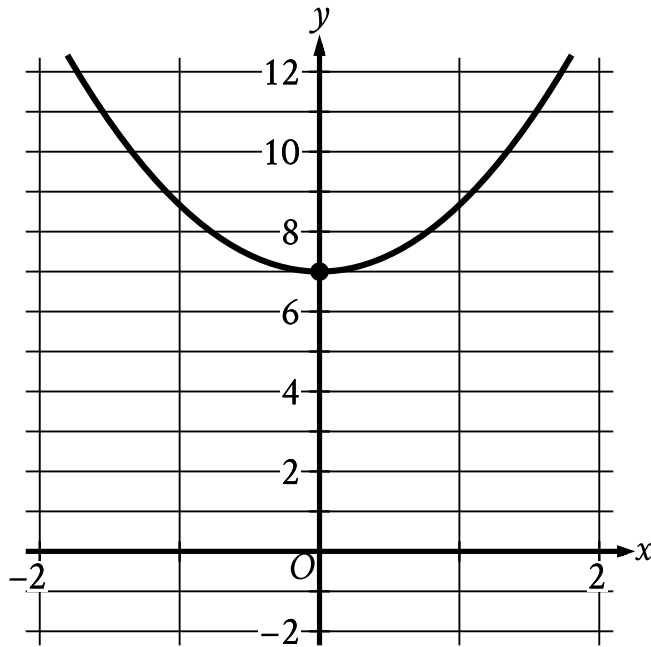
- A. 13,500
- B. 3,000
- C. 1,250
- D. 250



- The parabola opens upward.
- The parabola passes through the following points:
  - (-1, 5)
  - (0, 2)
  - (1, 5)

The graph of the quadratic function  $y = f(x)$  is shown. What is the vertex of the graph?

- A. 0, -2
- B. 0, -3
- C. 0, 2
- D. 0, 3



- The parabola opens upward.
- The vertex is at the point (0 comma 7).
- The parabola passes through the following points:
  - (negative 1 comma StartFraction 26 Over 3 EndFraction)
  - (0 comma 7)
  - (1 comma StartFraction 26 Over 3 EndFraction)

The parabola shown intersects the  $y$ -axis at the point  $x, y$ . What is the value of  $y$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |

|   |   |   |   |
|---|---|---|---|
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q10**

ADVANCED MATH

The function  $g$  is defined by  $gx = \sqrt{x} + 300$ . What is the value of  $gx$  when  $x = 81$ ?

- A. 9
- B. 300
- C. 309
- D. 381